

# HisabDo AI — Day 15

## Aarti — Ask HisabDo AI Response & Explanation Layer

Complete detailed specification + .NET/C# reference implementation

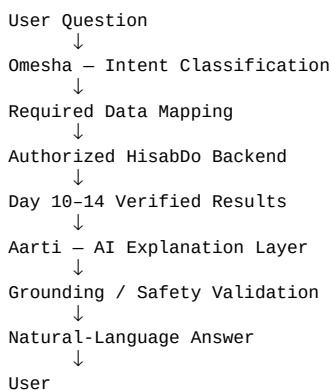
### 1. Module Objective

Aarti's responsibility is to convert verified HisabDo backend results into simple, natural-language answers. The AI explanation layer is not the source of truth. It must explain only the data supplied by the authorized backend and must safely communicate missing data, uncertainty, and limitations.

### 2. Exact 15-Task Coverage

1. Define AI response structure for Ask HisabDo.
2. Use only verified backend financial data.
3. Generate natural-language answers to user questions.
4. Explain financial values in simple language.
5. Explain spending patterns when requested.
6. Explain budgets and recommendations.
7. Explain financial health score.
8. Explain anomalies.
9. Explain cash-flow forecasts.
10. Explain limitations and uncertainty.
11. Ensure AI does not invent financial amounts.
12. Ensure AI does not answer using unsupported data.
13. Prepare sample Ask HisabDo AI responses.
14. Handle missing/insufficient data safely.
15. Define AI prompt and no-invention rules.

### 3. Correct Ask HisabDo Architecture



**Golden rule:** AI explains verified results; AI does not become the financial calculation engine.

### 4. AI Response Structure

| Section                     | Requirement                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------|
| 1. Direct Answer            | Answer the exact user question first, in simple language.                                   |
| 2. Key Verified Facts       | Mention only facts/values supplied by the backend.                                          |
| 3. Simple Explanation       | Explain what the verified value or pattern means.                                           |
| 4. Observation / Action     | Give an observation or action only when it is supported by verified backend results.        |
| 5. Limitation / Uncertainty | State insufficient history, missing budget, forecast confidence, or other known limitation. |

## 5. Verified Backend Data Rule

The backend must supply a controlled verified context. Depending on the intent, this can contain verified transaction totals, Day 11 spending patterns, Day 12 anomaly results, Day 13 forecasts, Day 14 budget recommendations, or the Day 10 financial health score. The AI must not query an unrelated source to fill missing HisabDo values.

## 6. Explanation Rules by Feature

| Feature                   | AI Explanation Rule                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Financial Values          | Explain income, expenses, totals, percentages, or balances exactly as provided. Do not recalculate a different number.       |
| Spending Patterns         | Explain top categories, increases/decreases, stable patterns, new categories, or recurring candidates only when relevant.    |
| Budgets & Recommendations | Explain existing budget status or verified recommended budget. Explain the reason supplied by the recommendation.            |
| Financial Health          | Explain the verified 0–100 score and six factor results from Day 10. Do not independently recalculate or alter the score.    |
| Anomalies                 | Explain verified anomaly type/severity and supplied transaction context. Do not label a user fraudulent or accuse anyone.    |
| Cash-Flow Forecast        | Explain verified expected income, expenses, savings, closing balance, confidence, and limitations from Day 13. Do not guess. |
| Limitations               | Clearly state insufficient history, missing budget, missing opening balance, low confidence, or other backend limitations.   |

## 7. No-Invention Rules

- Every financial amount in the answer must be traceable to verified backend context.
- Never invent an amount, percentage, score, date, category, transaction, budget, forecast, or recommendation.
- Never change a backend value because the AI thinks another value is more reasonable.
- Never calculate a new financial value inside the LLM when the backend has not supplied it.
- Never use another user's data.
- Never use outside financial data to fill a missing HisabDo value.
- Never turn an anomaly into a fraud/theft accusation without trusted evidence.
- Never hide a known limitation.
- If the backend does not support the requested answer, state that clearly.

## 8. Recommended AI System Prompt

You are Ask HisabDo AI, the explanation layer of HisabDo.

Answer the user's financial question using ONLY the verified financial context supplied by the authorized HisabDo backend.

Rules:

1. Explain verified values in simple language.
2. Never invent financial amounts, percentages, scores, dates, categories, transactions, forecasts, budgets, recommendations, or anomaly facts.
3. Never recalculate or alter a backend value.
4. If a required value is missing, do not guess.
5. Do not use outside data to fill missing HisabDo information.
6. Explain uncertainty and limitations supplied by the backend.
7. Answer only what the detected intent and verified context support.
8. Do not make fraud/theft/unauthorized-activity accusations unless the trusted backend explicitly provides that conclusion.
9. Keep the answer concise, clear, and useful.

Response format:

- Direct answer
- Key verified facts
- Simple explanation
- Supported observation/action
- Limitation when applicable

## 9. Sample Ask HisabDo AI Responses

### Income & Expense

Q: How much did I spend this month?

Your verified HisabDo data shows that you spent PKR 45,000 this month. This is the expense total for the selected period.

### Spending Pattern

Q: Where is most of my money going?

Food is your highest spending category in the verified data, accounting for 30% of total spending for this period.

### Budget

Q: Am I over my food budget?

Your verified budget data shows that Food is above its current budget. The backend reports 138% budget utilization for the selected period.

### Budget Recommendation

Q: What food budget do you recommend?

The verified recommendation engine suggests a Food budget of PKR 15,000 based on the available historical spending data. The recommendation is based on the period and data available to HisabDo.

### Financial Health

Q: Why is my financial health score 68?

Your verified Financial Health Score is 68/100. The score is based on the six configured factors, including saving behavior, expense control, budget control, cash flow, debt/udhaar, and expense growth.

### Anomaly

Q: Did I have any unusual expenses?

Yes. The verified anomaly engine identified an Unusual Amount anomaly for an expense in the supplied results. I can explain the recorded amount and severity shown by the backend.

### Cash-Flow Forecast

Q: What may my cash flow look like next month?

The verified forecast provides the expected income, expected expenses, expected savings, and closing balance for the forecast period. The forecast confidence and limitations should be considered when interpreting it.

### Missing Data

Q: What is my budget utilization?

I don't have enough verified budget data to answer that safely. Please make sure a budget exists for the requested period.

### Unsupported

Q: What will the stock market do tomorrow?

That is outside the supported Ask HisabDo financial data scope, so I can't provide a HisabDo-based answer from the available verified data.

## 10. Missing / Insufficient Data Handling

- Missing transactions → say that transaction data is unavailable.
- Insufficient historical months → explain that the requested pattern/forecast/recommendation needs more history.
- No existing budget → do not invent budget utilization.
- No opening balance → do not provide a verified closing balance.
- Missing anomaly result → do not create an anomaly.
- Missing health result → do not independently recreate the score in the LLM.
- Missing recommendation result → do not create a recommendation amount.
- Conflicting or unclear intent → request clarification rather than guessing.

## 11. .NET/C# Reference Implementation

The companion C# service receives a `VerifiedFinancialContext`, checks data sufficiency, builds the strict system/user prompts, calls an abstract LLM provider, and leaves a production validation boundary for numeric/entity grounding. The actual HisabDo repositories, authentication, Day 10–14 services, and selected LLM provider must be mapped during integration.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading;
using System.Threading.Tasks;
namespace HisabDo.AI.Day15;
/// <summary>
/// Day 15 – Aarti: Ask HisabDo AI Explanation Layer.
/// This layer converts VERIFIED backend data into natural language.
/// It must never calculate, alter, or invent financial values.
/// </summary>
public sealed record VerifiedFinancialContext
{
    public required string UserId { get; init; }
    public required string Intent { get; init; }
    public required string AnswerContext { get; init; }
    public IReadOnlyList<string> VerifiedFacts { get; init; } = Array.Empty<string>();
    public IReadOnlyList<string> Limitations { get; init; } = Array.Empty<string>();
    public bool HasSufficientData { get; init; }
}
public sealed record AskHisabDoAiResponse
{
    public required string Summary { get; init; }
    public required string Answer { get; init; }
    public IReadOnlyList<string> KeyFacts { get; init; } = Array.Empty<string>();
    public IReadOnlyList<string> Limitations { get; init; } = Array.Empty<string>();
    public bool UsedVerifiedDataOnly { get; init; }
}
public interface IAskHisabDoLlm
{
    Task<string> GenerateAsync(
        string systemPrompt,
        string userPrompt,
        CancellationToken cancellationToken = default);
}
public sealed class AskHisabDoAiExplanationService
{
    private readonly IAskHisabDoLlm _llm;
    public AskHisabDoAiExplanationService(IAskHisabDoLlm llm)
    {
        _llm = llm;
    }
    public async Task<AskHisabDoAiResponse> AnswerAsync(
        string userQuestion,
        VerifiedFinancialContext context,
        CancellationToken cancellationToken = default)
    {
        if (string.IsNullOrEmpty(userQuestion))
            throw new ArgumentException("Question is required.", nameof(userQuestion));
        if (context is null)
            throw new ArgumentNullException(nameof(context));
        if (string.IsNullOrEmpty(context.UserId))
            throw new InvalidOperationException("Authenticated UserId is required.");
        // Do not let the LLM answer if verified backend data is unavailable.
        if (!context.HasSufficientData)
```

```

{
    var limitation = context.Limitations.Count > 0
        ? string.Join(" ", context.Limitations)
        : "There is not enough verified HisabDo data to answer this question.";
    return new AskHisabDoAiResponse
    {
        Summary = "I need more verified data to answer this safely.",
        Answer = limitation,
        KeyFacts = context.VerifiedFacts,
        Limitations = context.Limitations,
        UsedVerifiedDataOnly = true
    };
}
var systemPrompt = BuildSystemPrompt();
var userPrompt = BuildUserPrompt(userQuestion, context);
var generated = await _llm.GenerateAsync(
    systemPrompt,
    userPrompt,
    cancellationTokens);
// Production recommendation:
// Validate the generated response against the verified facts before returning it.
// A response containing unsupported financial values must be rejected/regenerated.
var safeAnswer = SanitizeAgainstVerifiedFacts(generated, context);
return new AskHisabDoAiResponse
{
    Summary = BuildSafeSummary(context),
    Answer = safeAnswer,
    KeyFacts = context.VerifiedFacts,
    Limitations = context.Limitations,
    UsedVerifiedDataOnly = true
};
}
private static string BuildSystemPrompt() =>
"""
You are Ask HisabDo AI, the explanation layer of HisabDo.
ROLE:
- Answer the user's financial question in simple, clear language.
- Use ONLY the verified financial data supplied by the backend.
- Explain verified values; do not calculate new financial values.
- Do not invent amounts, percentages, dates, categories, scores, transactions,
  forecasts, budgets, anomaly severity, or recommendations.
- Do not use outside financial data to fill missing HisabDo values.
- If required data is missing or insufficient, say so clearly.
- Never claim fraud, theft, or unauthorized activity unless the trusted backend
  explicitly provides that conclusion.
- Respect the detected intent and answer only what the supplied verified data supports.
- If the question is unsupported, say that HisabDo cannot answer it from the available data.
- Keep financial explanations simple and practical.
RESPONSE STRUCTURE:
1. Direct answer
2. Key verified facts
3. Simple explanation
4. Action/observation only when supported by verified data
5. Limitation/uncertainty when applicable
NO-INVENTION RULE:
Every financial number in the answer must exist in the supplied verified backend context.
Do not create a replacement number if a value is missing.
""";
private static string BuildUserPrompt(
    string question,
    VerifiedFinancialContext context)
{
    var facts = context.VerifiedFacts.Count == 0
        ? "None"
        : string.Join("\n- ", context.VerifiedFacts);
    var limitations = context.Limitations.Count == 0
        ? "None"
        : string.Join("\n- ", context.Limitations);
    return $""""
USER QUESTION:
{question}
DETECTED INTENT:
{context.Intent}
VERIFIED BACKEND CONTEXT:
{context.AnswerContext}
VERIFIED FACTS:
- {facts}
LIMITATIONS:
- {limitations}
INSTRUCTIONS:
Answer the user using only the verified context above.
Do not invent or recalculate any financial number.
If the verified context does not support an answer, explicitly state the limitation.
""";
}
private static string BuildSafeSummary(VerifiedFinancialContext context) =>
    $"Answer generated for verified intent: {context.Intent}.";
private static string SanitizeAgainstVerifiedFacts(

```

```

        string generated,
        VerifiedFinancialContext context)
    {
        // Reference boundary only.
        // Production should implement structured output + numeric/entity validation.
        // If validation fails, reject the answer and regenerate with a stricter prompt.
        if (string.IsNullOrEmpty(generated))
            return "I could not generate an answer from the verified data.";
        return generated.Trim();
    }
}
/// <summary>
/// Optional deterministic guard for important numeric facts.
/// In production, prefer structured LLM output and compare extracted numeric tokens
/// against an allow-list generated from the verified backend result.
/// </summary>
public static class AiGroundingRules
{
    public static readonly string[] ForbiddenBehaviors =
    {
        "Invent a financial amount",
        "Change a backend percentage",
        "Recalculate a verified score",
        "Create a forecast without a verified forecast result",
        "Create a budget without a verified recommendation",
        "Claim an anomaly not present in backend results",
        "Use another user's financial data",
        "Use external data to fill missing HisabDo values",
        "Present uncertainty as certainty"
    };
}

```

## 12. Production Grounding Validation

For production, add structured LLM output and deterministic validation before returning the answer. Extract numeric/entity claims from the generated response and compare them with an allow-list built from verified backend facts. If the response contains unsupported financial values or unsupported entities, reject/regenerate it. This is an additional safety layer; it does not replace backend authorization or source-of-truth calculations.

## 13. Testing Scenarios

1. Correct answer for income total.
2. Correct answer for expense total.
3. Top spending category explanation.
4. Category increase/decrease explanation.
5. Existing budget explanation.
6. Budget recommendation explanation.
7. Financial health score explanation.
8. Anomaly explanation.
9. Cash-flow forecast explanation.
10. Insufficient history response.
11. Missing budget response.
12. Missing opening balance response.
13. Unsupported question response.
14. Ambiguous question response.
15. Prompt injection attempt asking AI to ignore verified data.
16. LLM attempts to change a verified amount.
17. LLM invents a percentage not present in context.
18. Cross-user data isolation test.

## 14. Assumptions & Limitations

The explanation layer assumes that Omesha's classifier has already identified the intent and that Jaffer/backend services provide authorized, verified data. The AI layer itself is not responsible for financial calculation. Natural-language quality depends on the selected LLM. Numeric grounding must be enforced in production. Multilingual support, especially Urdu/Roman Urdu, may require additional prompt and evaluation work.

## 15. Aarti Acceptance Checklist

- Response structure is finalized.
- Only verified backend data is accepted.
- All Day 10–14 feature explanations are covered.
- Simple natural-language answers are supported.
- No-invention rules are documented in the prompt.
- Missing/insufficient data is handled safely.
- Unsupported questions are not answered using invented data.
- Limitations and uncertainty are communicated.
- Sample responses are prepared.
- Grounding validation boundary is defined.
- C# service is ready for integration with the actual HisabDo backend and selected LLM.

## 16. Final Definition

Aarti's Day 15 Ask HisabDo AI module is the controlled natural-language explanation layer between verified HisabDo financial results and the user. It answers supported questions in simple language, explains spending, budgets, recommendations, health, anomalies, and forecasts, communicates limitations, and strictly prevents unsupported financial values from becoming AI-generated facts.